

Article

CropsDisNet: An AI-Based Platform for Disease Detection and Advancing On-Farm Privacy Solutions

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Abstract: Crop failure is defined as crop production that is significantly lower than anticipated, resulting from plants that are harmed, diseased, destroyed, or influenced by climatic circumstances. With the rise in global food security concern, the earliest detection of crop diseases has proven to be pivotal in agriculture industries to address the needs of the global food crisis and on-farm data protection, which can be met with a privacy-preserving deep learning model. However, deep learning seems to be a largely complex black box to interpret, necessitating a prerequisite for the groundwork of the model's interpretability. Considering this, the aim of this study was to follow up on the establishment of a robust deep learning custom model named CropsDisNet, evaluated on a large-scale dataset named "New Bangladeshi Crop Disease Dataset (corn, potato and wheat)", which contains a total of 8946 images. The integration of a differential privacy algorithm into our CropsDisNet model could establish the benefits of automated crop disease classification without compromising on-farm data privacy by reducing training data leakage. To classify corn, potato, and wheat leaf diseases, we used three representative CNN models for image classification (VGG16, Inception Resnet V2, Inception V3) along with our custom model, and the classification accuracy for these three different crops varied from 92.09% to 98.29%. In addition, demonstration of the model's interpretability gave us insight into our model's decision making and classification results, which can allow farmers to understand and take appropriate precautions in the event of early widespread harvest failure and food crises.

Keywords: Crops Disease Detection; custom CNN model; deep learning in agriculture; privacy preserve Net; privacy protection in agriculture



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1. Introduction

The agricultural revolution, a pivotal event that occurred around 8000 to 12,000 years ago, marked a fundamental movement in human history by enabling stable food production, thereby fostering the growth of civilizations [1]. This revolution, characterized by significant advancement in technologies and agricultural practices, played a crucial role in food security under the dramatic increase in global populations, from 1 billion in the 18th century to 7.2 billion currently [2]. Thereby, global food security remains a pressing concern, with recent data indicating that approximately 733.4 million people worldwide

were undernourished in 2023, accounting for 9.1% of the global population. This situation is exacerbated by climate change, population growth, and conflicts, which disrupt food production and distribution systems. Notably, 153 Nobel and World Food Prize laureates have called for substantial increases in agricultural research and innovation to prevent a looming global hunger crisis. These examples underscore the critical need for effective crop disease management to ensure food security [3].

Despite the existence of more than 50,000 consumable plant species, only a small number are useful to our diet [4]. Corn contributes to 19.5% of calorie consumption, while wheat accounts for 15%, and roots and tubers like potatoes contribute 5.3%. According to 2019 statistics, corn is the world's most productive staple crop, with a production of 1.1 billion tons, followed by wheat (765 million tons) and potatoes (370 million tons). Although global food production has increased tremendously since the green revolution in the 1950s, global food security and nutrition face severe challenges. Climate change has emerged as the major challenging factor that contributes to the increased level of stresses affecting the production of these staple cereals and tubers [5,6]. Stresses, arising from both abiotic and biotic factors, significantly impact crop yields. Among biotic factors, plant diseases, often triggered by unfavorable climatic conditions, such as temperatures and rainfall fluctuations, has become a critical threat to agricultural productivity.

In the realm of global agriculture, the impact of diseases on crop yield is profound, accounting for losses between 20 and 40%. On average, crop losses due to diseases range from 10.1% to 28.1% in wheat, 19.5 to 41.4% in corn, and 8.1 to 21% in potatoes [7]. Stem, stripe, and leaf rust are three types of wheat rust diseases caused by the Basidiomycete genus *Puccinia* species *P. graminis* f. sp. *tritici*, *P. striiformis* f. sp. *tritici*, and *P. triticina*, respectively [8]. Fungal diseases also significantly reduce farm profits in potato crops by severely compromising yield and quality. The two most destructive leaf diseases for potato crops are early blight, caused by *Alternaria solani*, and late blight, caused by *Phytophthora infestans* [9]. Corn, another staple, is affected by a myriad of leaf diseases, including *Curvularia* leaf spot, dwarf mosaic, grey leaf spot, northern leaf blight, brown spot, round spot, rust, and southern leaf blight, further underscoring the widespread impact of plant disease [10]. As the global population is projected to reach 10 billion by 2050, meeting the food demands of this burgeoning population will necessitate a 60% increase in food production. Addressing these agricultural challenges is not just an issue of crop management but a critical imperative for ensuring future food security on a global scale.

Crop diseases are typically classified based on the causative agents, including fungi, bacteria, viruses, nematodes, and parasitic plants. Each category encompasses various diseases that can affect different parts of the plant, such as roots, stems, leaves, or fruits. Understanding these classifications at early yielding stages is essential for developing targeted disease management strategies. The timely detection of disease incidence can facilitate targeted treatment approaches, thereby enhancing the sustainability of agricultural practices. For example, several economically important plants are susceptible to root rot diseases caused by the fungus *Rhizoctonia solani*. Consequently, it is vital to develop strategies to impede the spread of root rot disease immediately upon pathogen identification [11]. Similarly, fusarium head blight in wheat is a major concern due to its production of mycotoxin, which poses severe risks to human and animal health [12].

Henceforth, enhancing our understanding of global food security challenges, the classification of crop diseases, and their impact on agricultural production has been pivotal throughout the decades, and addressing these issues through automated identification using deep learning models has proven crucial to prevent future food crises and ensure a stable food supply for the growing global population. The integration of artificial intelligence (AI) in agriculture, particularly for disease detection in crops, has made significant strides,

as evidenced by various studies (Table 1) employing methodologies like convolutional neural networks (CNNs) and spectral analysis. These advanced AI-based technologies offer promising accuracy in identifying diseases in crops such as wheat, corn, potatoes, and more, with some models achieving accuracy ratings as high as 91% to 98.9% (Table 1). However, an often-overlooked aspect of this technological advancement is the privacy of on-farm data and the model's interpretability of engaged deep learning models for crop disease classification, which will be addressed in this novel study. The automated classification of crop diseases using deep learning models presents significant privacy challenges for on-farm data. Such data often include sensitive information, including geolocation, environmental conditions, and visual records of crops, which can inadvertently expose critical details about farm operations, production capacities, and proprietary agricultural practices. Large-scale datasets required for training and deploying deep learning models are often stored on cloud platforms, making them susceptible to breaches or unauthorized access. Another challenge lies in the absence of a unified regulatory framework governing the use and protection of agricultural data. Farmers frequently lack clarity about data ownership and may have limited control over how their information is used or shared, particularly in cross-border collaborations where varying privacy standards complicate compliance.

Table 1. State-of-the-art recognition of various plant diseases using deep learning and other conventional approaches.

Crops	Dataset Details	Approaches	Accuracy (%)	Reference
Potato	PlantVillage Dataset [2]	Fine-tuned VGG 16	97.9	[9]
Maize	500 images from PlantVillage dataset and Google websites (including different periods of occurrence of maize leaf diseases)	GoogleNet	98.9	[10]
		Cifar10	98.8	
Corn Apple Tomato Potato	PlantVillage Dataset [2] (comprising 21,978 images, four types of crops—corn, apple, tomato, and potato) along with manual image collection	✓ CNN-based proposed model Classification of crop diseases ✓ Single shot detector (identification and localization of the leaf)	96.9	[13]
Corn	PlantVillage Dataset	✓ CNN based proposed model	94.0	[14]
Wheat	Yellow-Rust-19, consisted of 15,000 images	✓ Yellow-Rust-Xception (Transfer Learning model based on Xception)	91.0	[15]
Potato	1125 ges from PlantVillage Dataset	✓ Classification with CNN	97.0	[16]
Wheat	Wheat-based dataset consisting of 1163 images collected by WARC (Walloon Agricultural Research Center)	✓ Mask R-CNN (segmentation and extraction)	93.5	[17]
		✓ GradCAM (visualization and interpretation)		
		✓ Accuracy tested using DenseNet121		

Since deep learning-based disease detection relies heavily on these data, ensuring their privacy is of the utmost importance since the potential risks associated with the breach of such data can be detrimental, resulting in various negative outcomes, including the disruption of crop yields, financial crises, compromised farming business strategies, unfair competition, and even threats to food and financial security [18]. Therefore, while acknowledging the benefits of AI in improving agricultural practices and disease management, it is crucial to address the privacy concerns associated with the use of on-farm

data. Developing robust data protection measures and privacy-preserving techniques in AI models is essential to protect this sensitive information from falling into the wrong hands. Doing so not only safeguards the interests of farmers and agricultural businesses but also contributes to the overall integrity and sustainability of the agricultural industry.

In recent years, the integration of differential privacy (DP) with deep learning models has garnered considerable attention due to its ability to protect sensitive user data while enhancing the precision and efficiency of these models. This integration is increasingly pertinent in sectors like healthcare, banking, and social media, where handling sensitive personal data is a critical concern. Recognizing the importance of data privacy, it is important to extend these measures to the agricultural sector. Particularly, in the context of crop disease classification using deep learning models, incorporating differential privacy is essential to prevent the leakage of confidential agricultural data. Our proposed methodology introduces the integration of differential privacy with the novel deep learning model, CropsDisNet. This model is designed to optimize utility while adhering strictly to privacy regulations. Utilizing the same datasets as other models, CropsDisNet demonstrates a consistent and accurate categorization of healthy and diseased leaf images, simultaneously addressing privacy concerns and minimizing the disclosure risk of confidential training data. This approach is instrumental in preserving the confidentiality of personal and intellectual property, ensuring efficient business operations—areas where conventional deep learning methods have fallen short. Furthermore, CropsDisNet places a high priority on model interpretability. This focus enhances the understanding of the model's decision-making process and the interpretation of the underlying data. Therefore, CropsDisNet represents a significant advancement in deep learning applications for agriculture, balancing the need for accuracy in disease detection with the imperative of maintaining data privacy.

2. Methodology

Figure 1 illustrates our recommended methodology for classifying crop diseases in wheat, potato, and corn leaves. Using the same dataset with proper data augmentation, our research method primarily compares the classification accuracy of our proposed model, CropsDisNet, with three existing pre-trained models (VGG16, Inception Resnet V2, and Inception V3). In order to justify the comprehension of our model's decision-making process, our suggested model promotes model interpretability using Grad-CAM while consistently classifying photos of healthy and diseased leaves. Furthermore, differential privacy in CropsDisNet addresses privacy issues, preventing the disclosure of private information from training data and opening up a whole new field for the classification of plant diseases to the best of our knowledge.

2.1. Data Preparation

This study used the image datasets derived from the leaves of three crops: corn, potato, and wheat. The dataset comprised a total of 8946 images, including 2942 images of wheat leaves, 2152 of potato leaves, and 3852 of corn leaves in a proportion of 7:2:1 for training, validation, and test sets. These images were further categorized into ten classes [19]. Comprehensive details of the dataset are presented in Table 2. Additionally, Figure 2 illustrates representative samples from each diseased crop leaf used for classifying crop diseases.

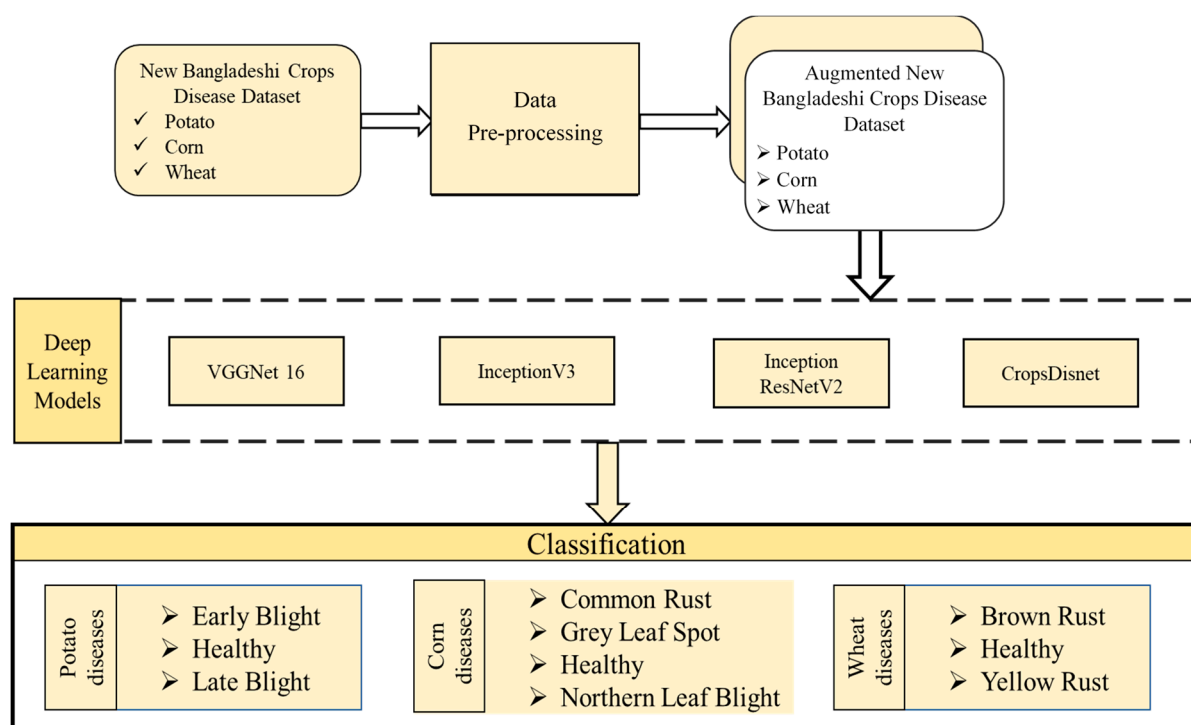


Figure 1. Demonstration of our proposed pipeline for the classification of crop diseases.

Table 2. Details of the New Bangladeshi Crop Disease Dataset.

Crop Name	Disease	Number of Images	Repository
Corn	Common Rust	1192	Kaggle (New Bangladeshi Crop Disease) [19]
	Gray Leaf Spot	513	
	Healthy	1162	
	Northern Leaf Blight	985	
Potato	Early Blight	1000	
	Healthy	152	
	Late Blight	1000	
Wheat	Brown Rust	902	
	Healthy	1116	
	Yellow Rust	924	

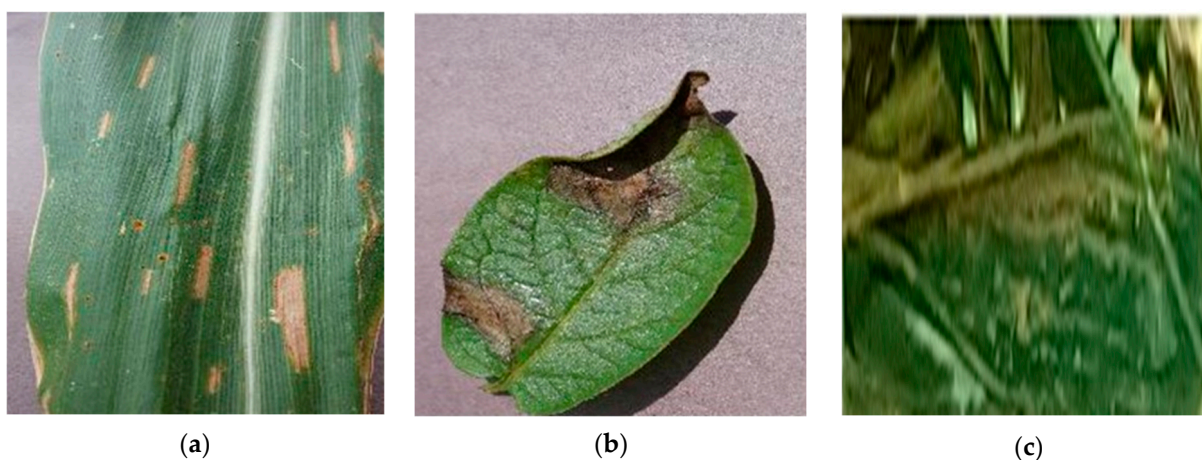


Figure 2. Images of (a) corn, (b) potato, and (c) wheat leaf.

The image datasets used in this study have undergone several stages of data augmentation and processing to enhance their utility. The primary aim of these techniques was to increase data size and introduce computational diversity, thereby mitigating the risk of

overfitting in the model and enhancing accuracy across varied image scenarios. The data augmentation techniques employed included scaling, cropping, padding, translation, rotation, flipping, and color augmentation. These methods collectively improved the quantity and quality of the data, making them more suitable for training the classification models. For instance, image scaling was tailored to match the input requirements of the neural networks. Every image was normalized to scale the data within a suitable range for the network. After conducting normalization on the image data, models were trained and evaluated. Additionally, contrast sketching techniques were utilized to enhance the visibility of low-contrast images. Uninformative areas within the images were removed through cropping, ensuring consistency in the intensity and size of the images. This consistency is important for leveraging the potential of pre-trained CNN models. Auto-cropping further refined the dataset, optimizing it for training efficacy. Color augmentation was another key step, wherein adjustments were made to the contrast, saturation, and brightness to enhance the overall image quality. Finally, all images were resized to a uniform dimension of 224×224 pixels. This comprehensive pre-processing and augmentation was instrumental in preparing the dataset (as shown in Figure 3) for efficient and effective use in the convolutional neural network.

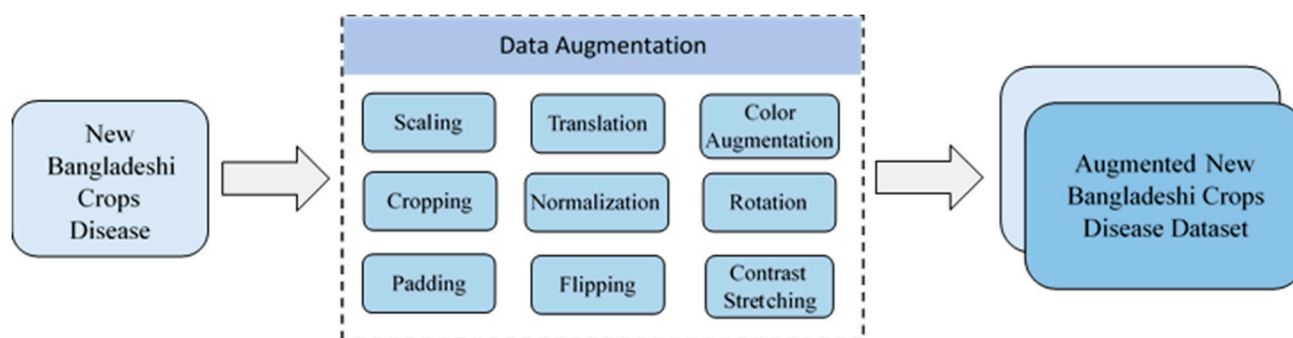


Figure 3. Preparation of dataset with data augmentation.

2.2. CNN Architectures in the Proposed Experimental Setup

This study employed deep learning using CNNs for image classification, enabling machines to identify and extract information from images. To train models to recognize target classes in images, labeled sample images were used. The CNN models operate on features extracted directly from the images, thereby eliminating the need for manual feature extraction. In this study, we compared our proposed CropsDisNet with three existing CNN architectures: VGG16, Inception Resnet V2, and Inception V3. Table 3 presents the training parameters for the four CNN models used in this study. This table includes information on the optimizer, base learning rate, momentum, learning decay rate, and train batch size for each model.

Table 3. The training parameters of the four CNN models involved.

Models	Optimizer	Base Learning Rate	Momentum	Learning Decay Rate	Train Batch Size
VGG-16	SGD	$1e^{-4}$	0.9	$1e^{-6}$	16
Inception v3	SGD	$1e^{-4}$	0.9	$1e^{-6}$	16
Inception_Resnet-V2	SGD	$1e^{-4}$	0.9	$1e^{-6}$	16
CropsDisNet	Differentially Private Gradient Descent	$1e^{-4}$	0.9	$1e^{-6}$	16

VGGNet-16, proposed by Karen Simonyan and Andrew Zisserman in 2014, achieved high accuracy in image classification and object localization in the ILSVRC challenge [20]. This model, with its 16 convolutional layers, is popular for image feature extraction. It has several filters with 3×3 convolutions, like AlexNet [21]. On 4 GPUs, it can be trained in two to three weeks.

Inception V3, an improvement over the earlier Inception models, was developed by a team at Google [22]. It features a 42-layer architecture with significant enhancements, including factorization into smaller convolutions and effective grid size reduction.

Inception-ResNet-v2, a hybrid model, combines residual connections with Inception-style networks to enhance recognition performance [23]. It has a total of 164 layers and demonstrates significant performance improvements over traditional Inception models.

Our custom model, CropsDisNet, features nine convolutional layers, three max-pooling layers, and a sequence of two fully connected layers. The network architecture begins with an input layer processing 224×224 pixel images composed of three color channels. Subsequently, three convolutional blocks are employed, each comprising three convolutional layers with 64, 128, and 256 filters, respectively, followed by max pooling and dropout. The final convolutional block's output is transformed into a one-dimensional feature vector, subsequently fed into a fully connected layer containing 4096 neurons, employing a ReLU activation function, and followed by dropout. An additional, identical, fully connected layer is appended, followed by an output layer consisting of three neurons to perform the desired three-class classification task. Lastly, a softmax activation function is utilized in the output layer to generate a probability distribution for each class. The complete architectural design of the CropsDisNet model is illustrated in Figure 4.

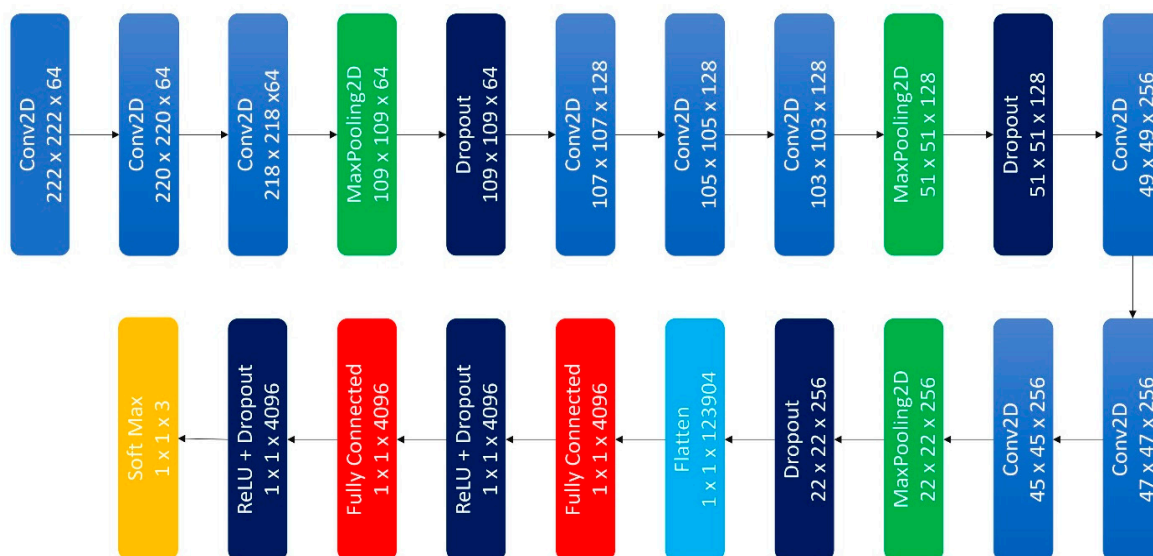


Figure 4. CNN architectures of our custom model CropsDisNet.

CropsDisNet incorporates ReLU (Rectified Linear Unit) activations to mitigate overfitting in the training dataset. For optimization, CropsDisNet employs a differentially private gradient descent. This involves clipping gradients on a per-example basis to limit their sensitivity and adding spherical Gaussian noise to maintain the indistinguishability required for differential privacy (DP). The ReLU function adds non-linearity to the network, and the softmax activation function generates output values between [0] and [1]. Categori-

cal cross-entropy was used as the loss function for this model. Equation (1) provides an explanation of the softmax output [24].

$$\text{Softmax}(y = j | \theta^i) = \frac{e^{\theta^i}}{\sum_{j=0}^k e^{\theta^i}} \quad (1)$$

where $\theta = w_0 f_0 + w_1 f_1 + \dots + w_k f_k = \sum_{i=0}^k w_i f_i = W^T F$.

Here, the categorical variables are represented as binary vectors by the one-hot encoded matrix θ . This function determines whether a group of features, f , belongs to a class, j , and the optimal output is the exponential of the input parameter divided by the sum of all parameters. Categorical cross-entropy is the loss function. The loss function is utilized when an example can only be classified under one of the potential classes, as shown in Equation (2). A detailed layout design of our proposed CropsDisNet's architecture is demonstrated in Table 4.

$$\text{Loss} = - \sum_{i=1}^{\text{output size}} y_i \times \log y_i \quad (2)$$

where $y_i = i^{\text{th}}$ scalar value for the output of the model.

Table 4. Detailed architecture layout of the proposed model CropsDisNet.

Layer	Input	Operation	Parameter Size	Strides	Output
1	$224 \times 224 \times 3$	Conv2D + Relu	$64 \times 3 \times 3$	1	$222 \times 222 \times 64$
2	$222 \times 222 \times 64$	Conv2D + Relu	$64 \times 3 \times 3$	1	$220 \times 220 \times 64$
3	$220 \times 220 \times 64$	Conv2D + Relu	$64 \times 3 \times 3$	1	$218 \times 218 \times 64$
4	$218 \times 218 \times 64$	Maxpool2D	2×2	2	$109 \times 109 \times 64$
5	$109 \times 109 \times 64$	Dropout	0.3	-	$109 \times 109 \times 64$
6	$109 \times 109 \times 64$	Conv2D + Relu	$128 \times 3 \times 3$	1	$107 \times 107 \times 128$
7	$107 \times 107 \times 128$	Conv2D + Relu	$128 \times 3 \times 3$	1	$105 \times 105 \times 128$
8	$105 \times 105 \times 128$	Conv2D + Relu	$128 \times 3 \times 3$	1	$103 \times 103 \times 128$
9	$103 \times 103 \times 128$	Maxpool2D	2×2	2	$51 \times 51 \times 128$
10	$51 \times 51 \times 128$	Dropout	0.3	-	$51 \times 51 \times 128$
11	$51 \times 51 \times 128$	Conv2D + Relu	$256 \times 3 \times 3$	1	$49 \times 49 \times 256$
12	$49 \times 49 \times 256$	Conv2D + Relu	$256 \times 3 \times 3$	1	$47 \times 47 \times 256$
13	$47 \times 47 \times 256$	Conv2D + Relu	$256 \times 3 \times 3$	1	$45 \times 45 \times 256$
14	$45 \times 45 \times 256$	Maxpool2D	2×2	2	$22 \times 22 \times 256$
15	$22 \times 22 \times 256$	Dropout	0.3	-	$22 \times 22 \times 256$
16	$22 \times 22 \times 256$	flatten	123,904	-	$123,904 \times 1$
17	$123,904 \times 1$	Dense + Relu	4096	-	4096×1
18	4096×1	Dropout	0.5	-	4096×1
19	4096×1	Dense + Relu	4096	-	4096×1
20	4096×1	Dropout	0.5	-	4096×1
21	4096×1	Dense + Softmax	3	-	3×1

This comprehensive approach to CNN architecture and optimization in our study offers a robust framework for image classification, particularly in the context of identifying crop diseases, while ensuring the privacy and security of the training data.

2.3. Integration of Differential Privacy in CropsDisNet: Implementation and Application

In the development of CropsDisNet, a key concern was addressing the privacy of sensitive data within the utilized datasets. Given the expansive availability of datasets in deep

learning, many of those contain confidential information that necessitates rigorous protection. We constructed our CropsDisNet utilizing commonly accepted privacy-preserving techniques known as differential privacy (DP) in order to prevent this unwanted situation [25]. Differential privacy is a renowned method for safeguarding sensitive data, and its integration into CropsDisNet is illustrated in Figure 5. This integration involved the implementation of the differentially private gradient descent algorithm, combined with meticulous hyperparameter tuning. These techniques collectively ensure the confidentiality of data during the training process. The primary goal of using DP in this context is to evaluate and minimize privacy loss effectively, thus maintaining the integrity of the data. Additionally, we employed empirical methods for the quantification of privacy loss. This involves the utilization of computational strategies to provide an asymptotical estimation of privacy loss [26,27]. Such an empirical approach enables the accurate measurement and control of privacy risks, ensuring that CropsDisNet remains both a robust and secure tool for crop disease classification. By balancing the need for efficient data processing with stringent privacy safeguards, our model represents a significant advancement in the responsible use of AI in agriculture.

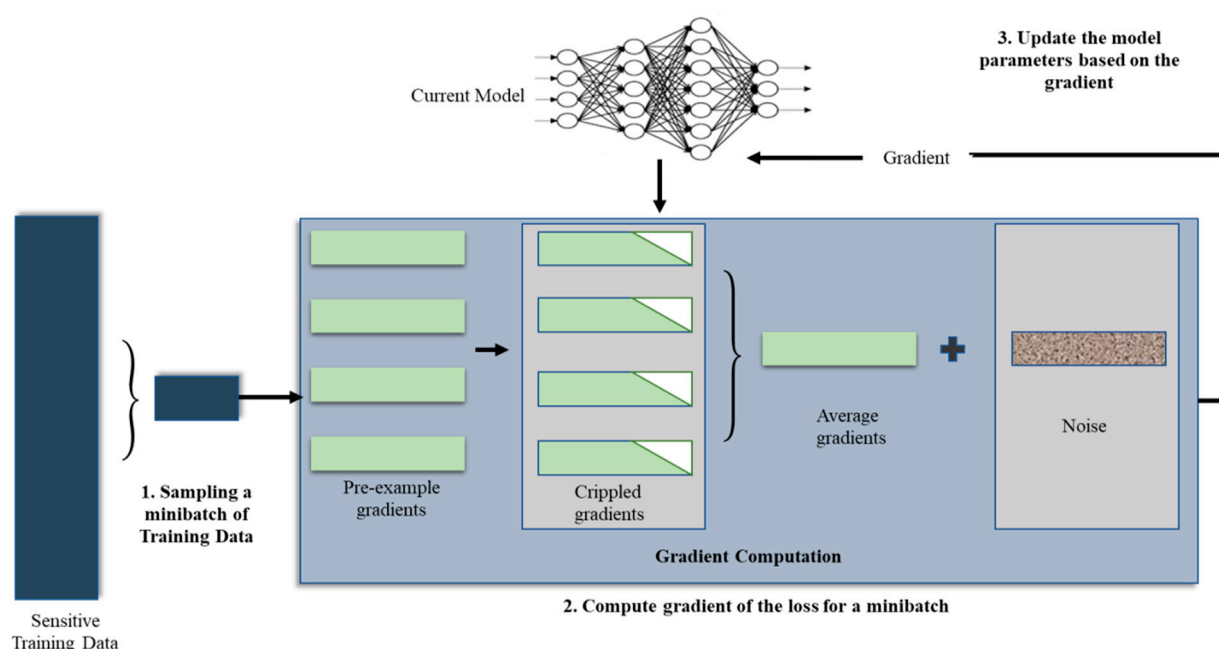


Figure 5. Implementation of differential privacy in our custom CropsDisNet model.

Each gradient's sensitivity must be regulated during the development of a differentially private model in order to regulate the gradient computations implemented on the parameters of CropsDisNet. The optimizer computes (Table 5) the average model error on these examples that are differentiated and a gradient vector on the model and modifies them at each training point.

Table 5. Detailed information on hyperparameters of our proposed CropsDisNet.

Differential Private Hyper-Parameter	Value
L2_norm_clip	1.5
Noise_multiplier	5
Num_microbatches	250
Learning rate	0.0009

The next step is to measure the gradients of the loss for the training dataset after clipping and adding Gaussian noise. Furthermore, the optimizer's sensitivity to specific training points is constrained by clipping each gradient generated on each training point using the l2 norm clip.

While the training points allow for clipping gradients on a per-sample basis, the noise multiplier tends to quantify the sampling and adding of noise to the gradients by five times in this model. In order to update the model's parameters sequentially, the gradient needs to be deducted first and then multiplied by the learning rate that needs to be performed. Using this, the final privacy loss during the training of any dataset by deep learning methods is reduced at a low cost in terms of software complexity, training proficiency, and model quality [28]. In coherence with this differential private strategy, deep neural networks may be achieved at a lower cost regarding software complexity, training skill, and model quality while reducing the overall loss of differential privacy.

2.4. Visualizing Interpretability of CropsDisNet with Grad-CAM

Grad-CAM, which stands for gradient-weighted class activation mapping, is a technique that we applied to support CropDisNet, our own model, to validate the model's interpretability. Grad-CAM is solely employed with our custom model, as the other three models have previously been established, allowing us to defend our model's decision-making process, as well as recognize the underlying model's explainability, thus leading us to potential insight into the model's performance. It uses gradients between the classification score and the final convolutional feature map, as seen in Figure 6. This method indicates the areas of an input image that significantly influence the classification score. The final score depends on the input data, mostly in the locations where this gradient is large. Although conventional CNNs excel in classification tasks, they often operate as "black boxes" due to the complexity of their multi-layered structures and numerous parameters. This opacity can undermine trust in the model's predictions. Grad-CAM addresses this issue by generating a heatmap that highlights the relevant regions of an image influencing the final convolutional layer's output during classification tasks. This feature is particularly important in preventing misclassification and offers a clear visualization of the decision-making process without necessitating alterations to the existing network architecture.

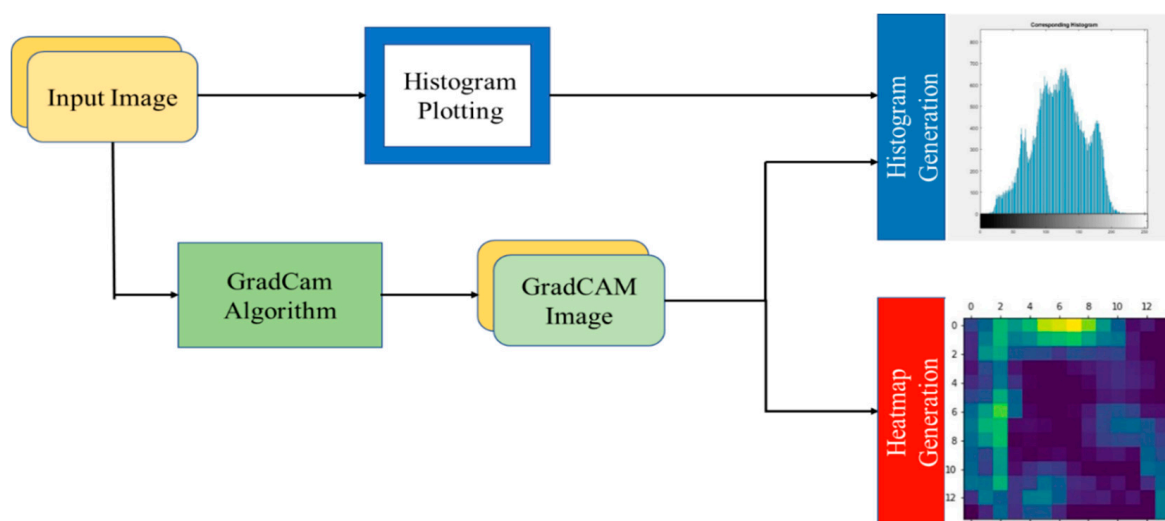


Figure 6. Pipeline for evaluating model's interpretability using Grad-CAM algorithm with their corresponding histograms.

To employ Grad-CAM with our CropsDisNet model, an image is first processed to produce a raw classification score for a given category. During this process, gradients for all classes are set to zero, except for the target class (the specific plant disease), which is assigned a value of 1. This targeted signal is then backpropagated to the rectified convolutional features to produce what is commonly referred to as the Grad-CAM localization heatmap. This blue heatmap explicitly illustrates the areas the model focuses on. This grad-CAM technique (Figure 6) is applied to the enhanced input image, which results in the generation of both heatmaps and histograms. These visual tools provide a transparent explanation of how the model formulates its predictions, thereby significantly enhancing the interpretability of the CropsDisNet model. This methodological approach not only reinforces the reliability of the model's outputs but also contributes to a deeper understanding of its decision-making process.

3. Results

The experimental findings of three pre-trained models and CropsDisNet across the dataset are compared in this section, taking into account a number of performance matrices including accuracy, loss, precision, recall, F1-Score, and AUC Score. Additionally, the model's performance is shown for three distinct crop disease classification types for CropsDisNet in terms of ROC curve and the confusion matrix. Moreover, GradCAM is used to assess CropsDisNet's interpretability by producing heatmaps that show the regions that are crucial to determining whether a leaf is healthy or diseased. Figure 7 illustrates the accuracy and AUC Score for the overall model, using a dataset that includes three different pre-trained models along with our custom model CropsDisNet. The New Bangladeshi Crop Disease Dataset was subjected to each of the three crop diseases that follow corn, potato, and wheat. Both VGG-16 and Inception_ResNetV2 achieved a maximum accuracy of 97% in the classification of corn diseases, whereas Inception_ResNetV2 had the highest maximum AUC score of 0.9969. Additionally, CropsDisNet showed the highest accuracy of 98.6% in the classification of potato diseases, whereas Inception_ResNetV2 once again demonstrated the highest AUC Score of 0.9993. Finally, to classify wheat diseases, VGG16 demonstrated the highest accuracy of 98%, as well as the highest AUC Score of 0.9991.

3.1. Performance Matrices of Engaged Models

Table 6 demonstrates a comparative analysis of the experimental results from different models where the performance matrices of three pre-trained models in coherence with CropsDisNet over the existing dataset had led so far. Different performance matrices, including precision, recall, and F₁-score, have been plotted for one versus rest (one versus all) approach for each of the classes, whereas overall performance comparison has been plotted in Figure 7.

Table 6 shows the test performance matrices of three other pre-trained models along with our custom model CropDisNet. Each has been applied to the New Bangladeshi Crops Disease Dataset, including three crop diseases following corn, potato, and wheat.

In terms of corn disease classification, maximum accuracy scored 97% from both VGG-16 and Inception_ResNetV2, which also provided F₁ scores of 0.9968 and 0.9969, respectively, whereas InceptionV3, along with our proposed CropsDisNet, achieved an accuracy score of 95% for corn diseases regarding common rust, gray leaf spot, and healthy classes. In the case of potato disease classifications, including the common rust, gray leaf spot, healthy, and northern leaf blight classes, our proposed CropsDisNet emerged with the highest accuracy of 98.6% and F₁ score of 0.9993, followed by the other two models (VGG-16 and Inception_ResNetV2) with an accuracy score of 98%, although InceptionV3 demonstrated less accuracy of 92% compared to the other three models. Lastly, VGG16

achieved the highest accuracy of 98% and an F_1 score of 0.9983 for the classification of wheat diseases corresponding to the brown rust, healthy, and yellow rust classes. In contrast, our proposed CropsDisNet also showed promising results close to VGG16 with an accuracy of 97%, which was also matched by Inception_ResNetV2. Like before, InceptionV3 achieved the least accuracy at 94%.

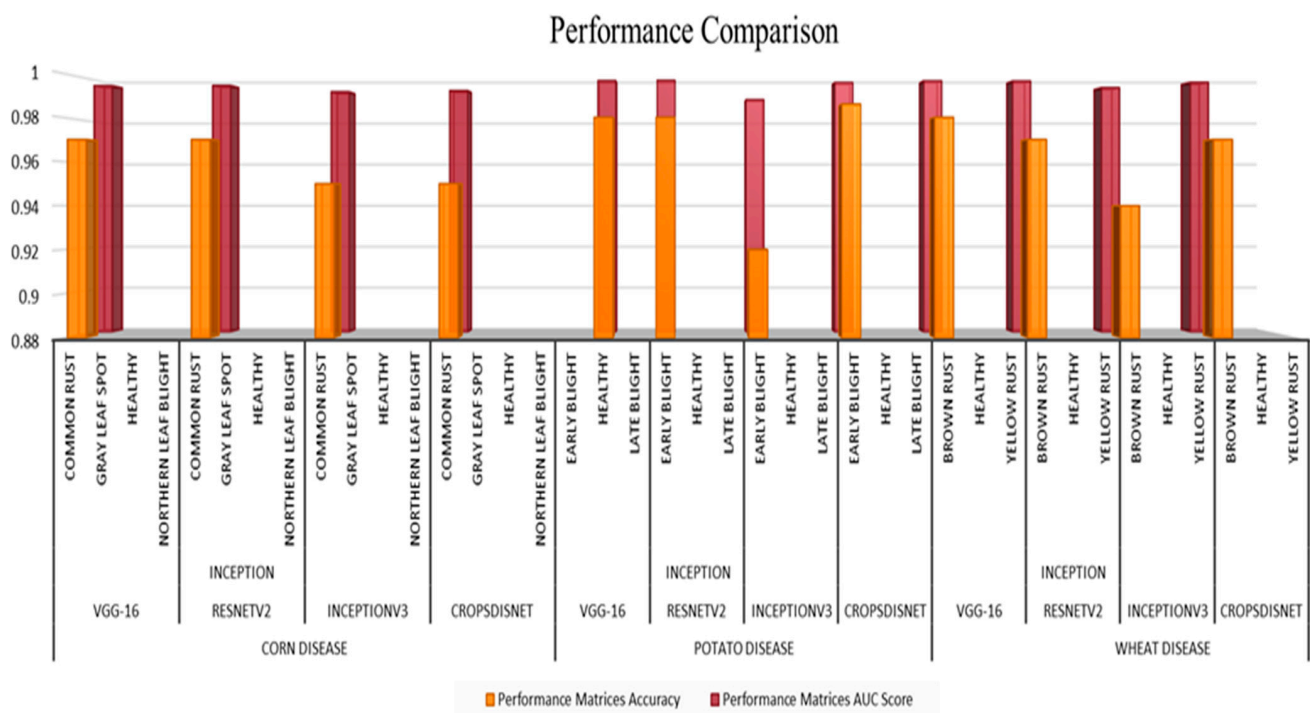


Figure 7. Accuracy and AUC for the overall model on datasets.

Table 6. Test performance matrices of four engaged models on the New Bangladeshi Crop Disease Dataset.

Part of the Dataset	Engaged Deep Learning Models	Class of the Diseases	Performance Matrices				
			Precision	Recall	F1-Score	Accuracy	AUC
Corn Disease	VGG-16	Common Rust	0.99	1.00	1.00	0.97	0.9968
		Gray Leaf Spot	0.86	0.94	0.90		
		Healthy	1.00	0.98	0.99		
		Northern Leaf Blight	0.97	0.93	0.95		
	Inception_ResNetV2	Common Rust	1.00	1.00	1.00	0.97	0.9969
		Gray Leaf Spot	0.90	0.85	0.87		
		Healthy	1.00	1.00	1.00		
		Northern Leaf Blight	0.92	0.95	0.94		
	InceptionV3	Common Rust	0.99	1.00	1.00	0.95	0.9939
		Gray Leaf Spot	0.84	0.83	0.83		
		Healthy	0.97	1.00	0.99		
		Northern Leaf Blight	0.93	0.90	0.91		
	CropsDisNet	Common Rust	1.00	1.00	1.00	0.95	0.9945
		Gray Leaf Spot	0.77	0.92	0.84		
		Healthy	1.00	1.00	1.00		
		Northern Leaf Blight	0.96	0.86	0.90		

Table 6. *Cont.*

Part of the Dataset	Engaged Deep Learning Models	Class of the Diseases	Performance Matrices				
			Precision	Recall	F1-Score	Accuracy	AUC
Potato Disease	VGG-16	Early Blight	1.00	0.99	0.99	0.98	0.9992
		Healthy	0.87	0.87	0.87		
		Late Blight	0.97	0.98	0.98		
	Inception_ResNetV2	Early Blight	0.99	0.99	0.99	0.98	0.9993
		Healthy	1.00	0.87	0.93		
		Late Blight	0.97	0.99	0.98		
	InceptionV3	Early Blight	0.93	0.98	0.96	0.92	0.9902
		Healthy	0.69	0.73	0.71		
		Late Blight	0.95	0.89	0.92		
	CropsDisNet	Early Blight	0.99	0.98	0.98	0.986	0.9983
		Healthy	0.93	0.87	0.90		
		Late Blight	0.96	0.98	0.97		
Wheat Disease	VGG-16	Brown Rust	0.95	1.00	0.97	0.98	0.9991
		Healthy	1.00	1.00	1.00		
		Yellow Rust	1.00	0.95	0.97		
	Inception_ResNetV2	Brown Rust	0.94	0.99	0.96	0.97	0.9990
		Healthy	0.99	0.98	0.99		
		Yellow Rust	0.99	0.95	0.97		
	InceptionV3	Brown Rust	0.86	0.99	0.92	0.94	0.9959
		Healthy	0.98	0.95	0.97		
		Yellow Rust	1.00	0.88	0.94		
	CropsDisNet	Brown Rust	0.92	1.00	0.96	0.97	0.9984
		Healthy	0.98	1.00	0.99		
		Yellow Rust	1.00	0.89	0.94		

Figure 8 shows the model's accuracy, the model's loss, and the ROC curve for the classification of diseases for three types of crops using our custom CropsDisNet model. Here, the blue line represents the training curve, whereas the orange one is for the validation curve. Both the training and validation accuracy in each of the model's accuracy diagrams varied in the range of 0.3–1.0. In the case of the model loss curve, the training and validation curve did not completely converge after 100 epochs except for the model loss of wheat disease classification, where both the training and validation curve potentially appeared to be diverged from each other. In the case of ROC curves, the horizontal axis represents the false positive rate, whereas the vertical axis represents the true positive rate. The values here varied in the range from greater than 0.99 to 1.00, where each of the classes for different crops included here properly converged to one point at the end.

Figure 9 concatenates the dynamic performance of the model CropsDisNet, which is depicted by a confusion matrix, one for each type of crop disease classification. The confusion matrix explicitly demonstrates the overall efficiency of the proposed model by asserting a very low number of false negatives and false positives while pushing for the numbers of true detections to be much higher. This results in excellent accuracy and F_1 scores and, thus, a feasibly accurate classification for each crop disease (potato, corn, and wheat) that are thoroughly described in previous discussions. For potato disease classification, 98 images were recognized as both true and predicted, labeled positively as early blight and late blight, respectively. Moving to the confusion matrix of corn disease classification, the numbers of true positives are 119, 48, 116, and 85, which were found

for common rust, gray leaf spot, healthy corn, and northern leaf blight, both at true and predicted labels. Lastly, true positives in terms of wheat disease classification were 90 for brown rust, 111 for healthy plants, and 82 for yellow rust in wheat disease image samples, whereas, like the other two (corn and potato), the no. of false positives and false negatives was deemed to be very low.

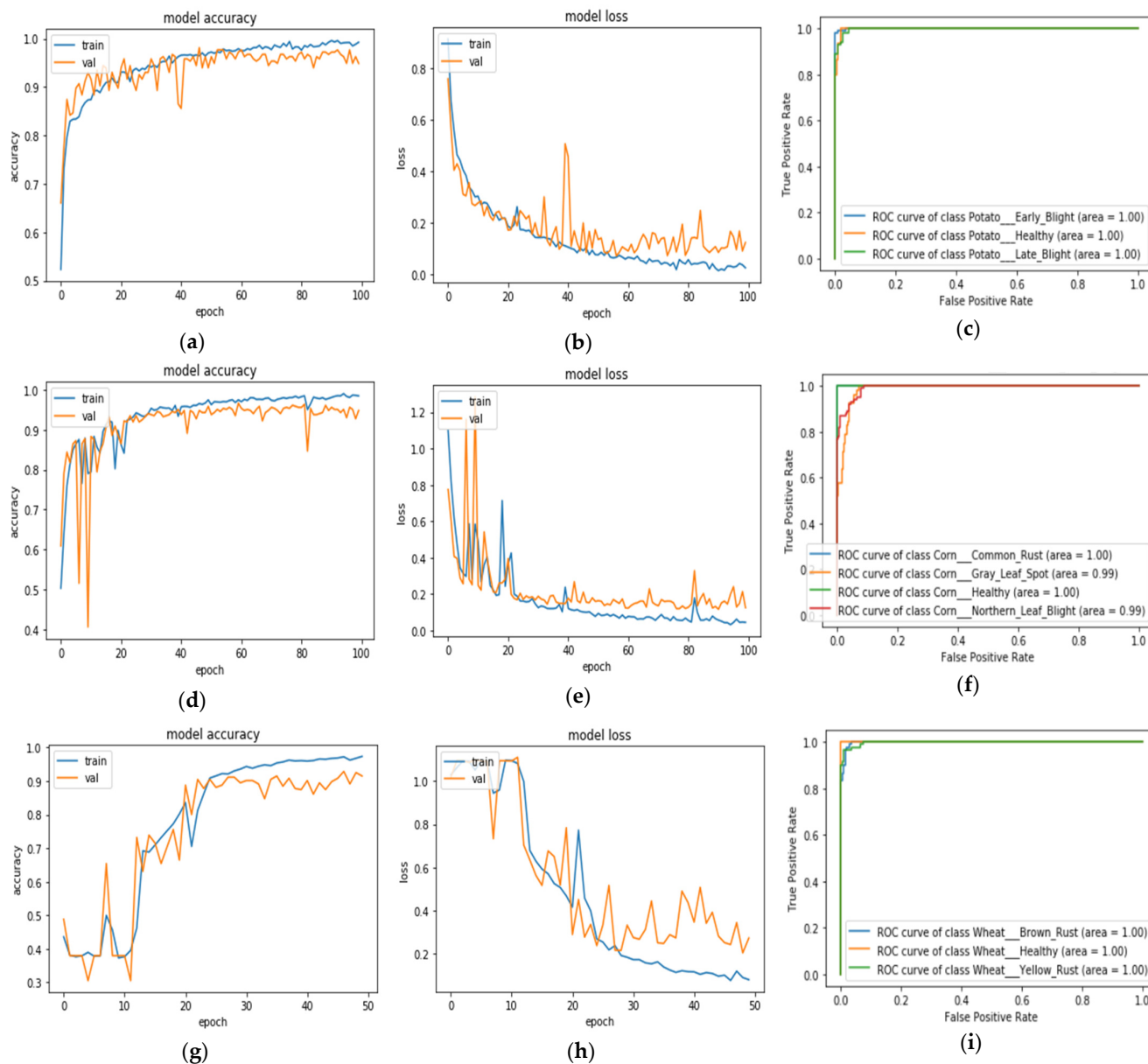


Figure 8. CropsDisNet's (a) training and validation accuracy, (b) training and validation loss curve, and (c) ROC Plot for potato disease classification; (d) training and validation accuracy, (e) training and validation loss curve for corn disease classification, and (f) ROC Plot for corn disease classification; (g) training and validation accuracy, (h) training and validation loss curve for wheat disease classification, and (i) ROC Plot for wheat disease classification.

3.2. Interpretation of CropsDisNet on the Dataset

Model interpretability is the focal point of this section as one of our main contributions to this study since it helps to better understand how the model was able to produce feasible predictions for the classification of different crop diseases. The GradCAM algorithm was used to assess our model CropsDisNet's interpretability and three other pre-trained models to meet this purpose. GradCAM generates heatmaps that highlight salient regions within

an image and provide insights into the areas to determine the healthy or diseased state. Figures 10a, 11a and 12a depict original images from each of the crops (potato, corn, and wheat, respectively) in coherence with their corresponding histograms, as shown in Figures 10b, 11b and 12b. In contrast, the GradCAM image from the original image is demonstrated in Figures 10c, 11c and 12c, respectively, whereas Figures 10d, 11d and 12d show each of the respective corresponding histograms.

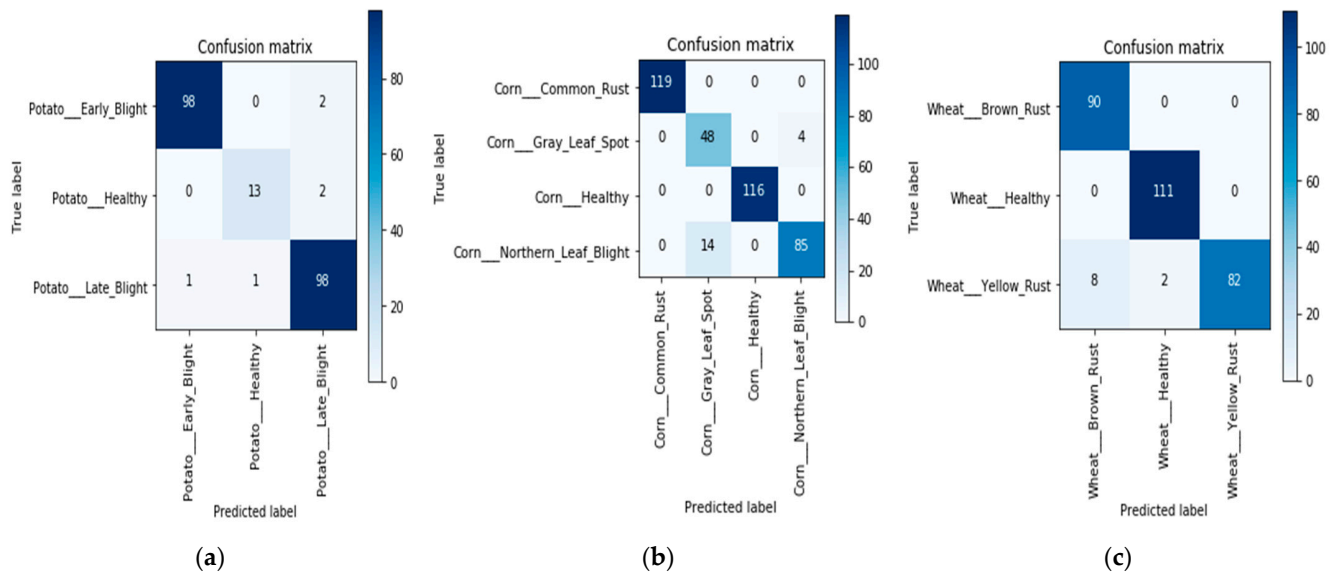


Figure 9. CropsDisNet’s confusion matrix for (a) potato disease classification (three classes), (b) corn disease classification (four classes), and (c) wheat disease classification (three classes) from the New Bangladeshi Crop Disease dataset.

Figures 10–12 illustrate input images of potato, corn, and wheat, respectively, along with their GRADCAM images with generated heatmaps and all of their corresponding histograms; the pixel intensity distribution of both the input image and its respective GRADCAM image are shown. The X-axis represents pixel intensity values, while the Y-axis indicates the number of pixels. Comparing both histograms, we observe shifts in pixel counts, enabling meaningful interpretations. Our application involves employing deep learning models for plant disease diagnosis, with the integration of the GRADCAM algorithm to highlight salient regions in the input image via heatmaps, facilitating enhanced interpretation. The heatmap reveals significant features recognized by the model, guiding its valuable decision-making process.

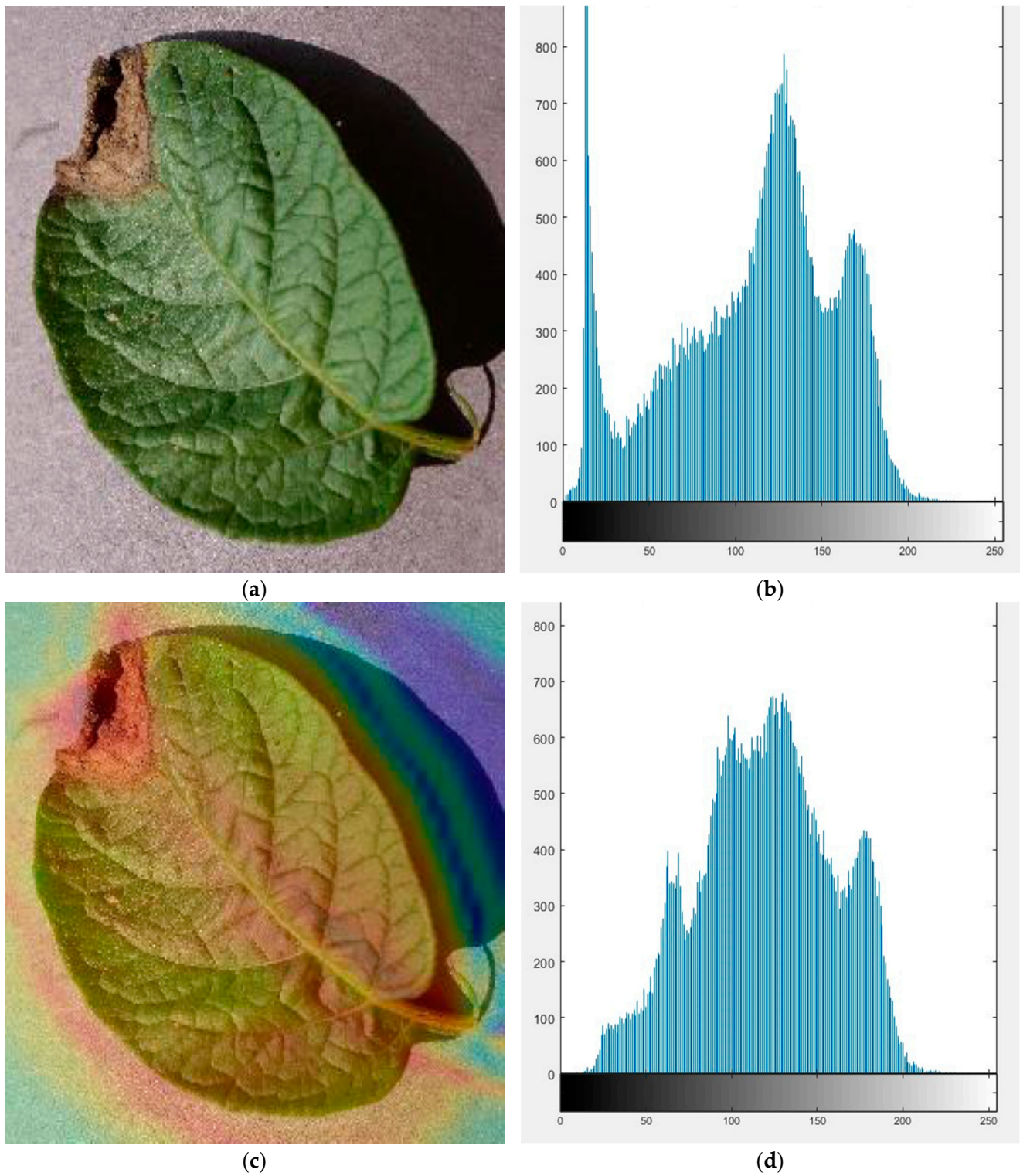


Figure 10. (a) Diseased potato image, (b) its corresponding histogram, (c) GRAD-CAM image of that diseased potato image, and (d) corresponding histogram of that GRAD-CAM image.

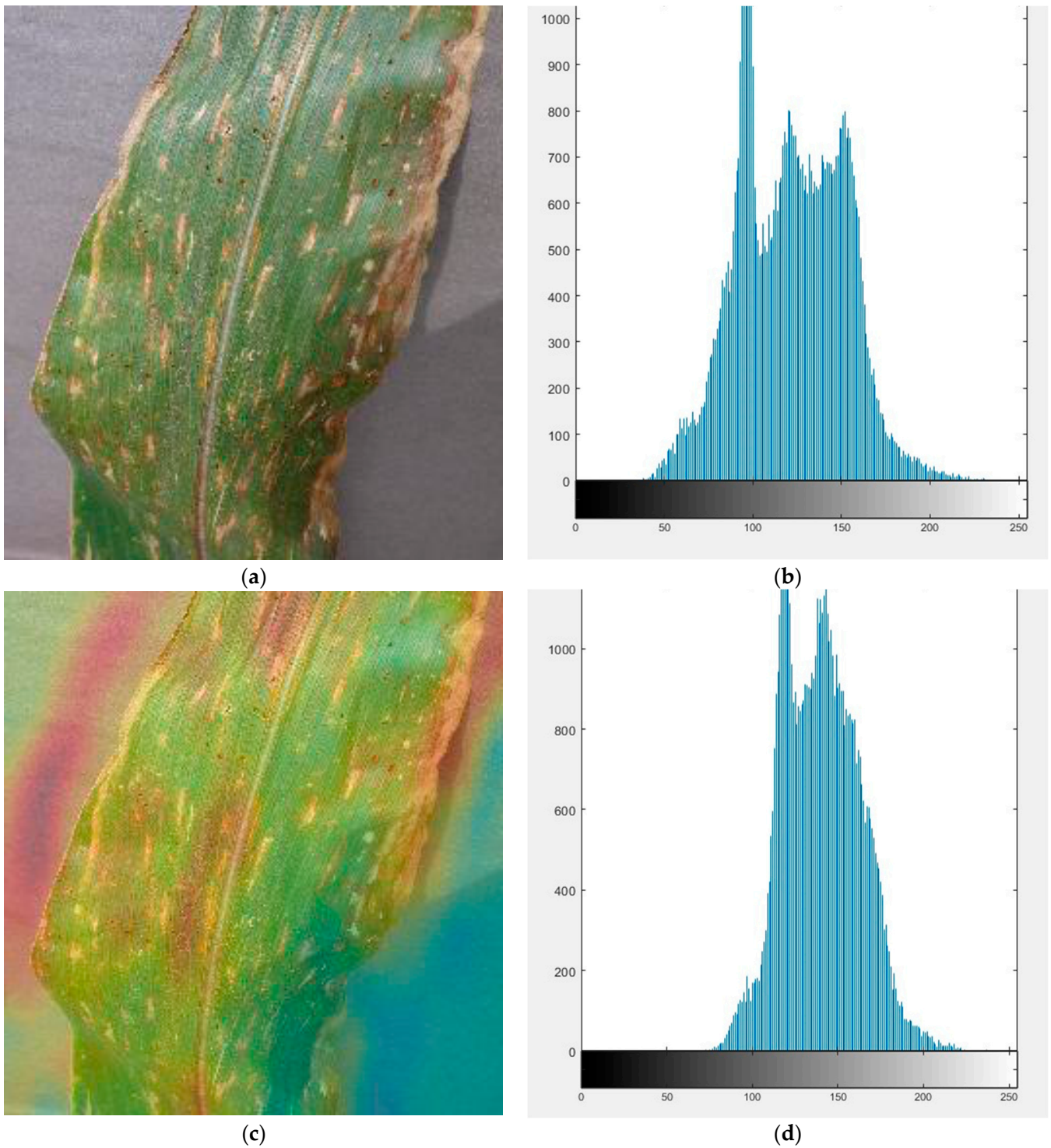


Figure 11. (a) Diseased corn image, (b) its corresponding histogram, (c) GRAD-CAM image of that diseased corn image, and (d) corresponding histogram of that GRAD-CAM image.

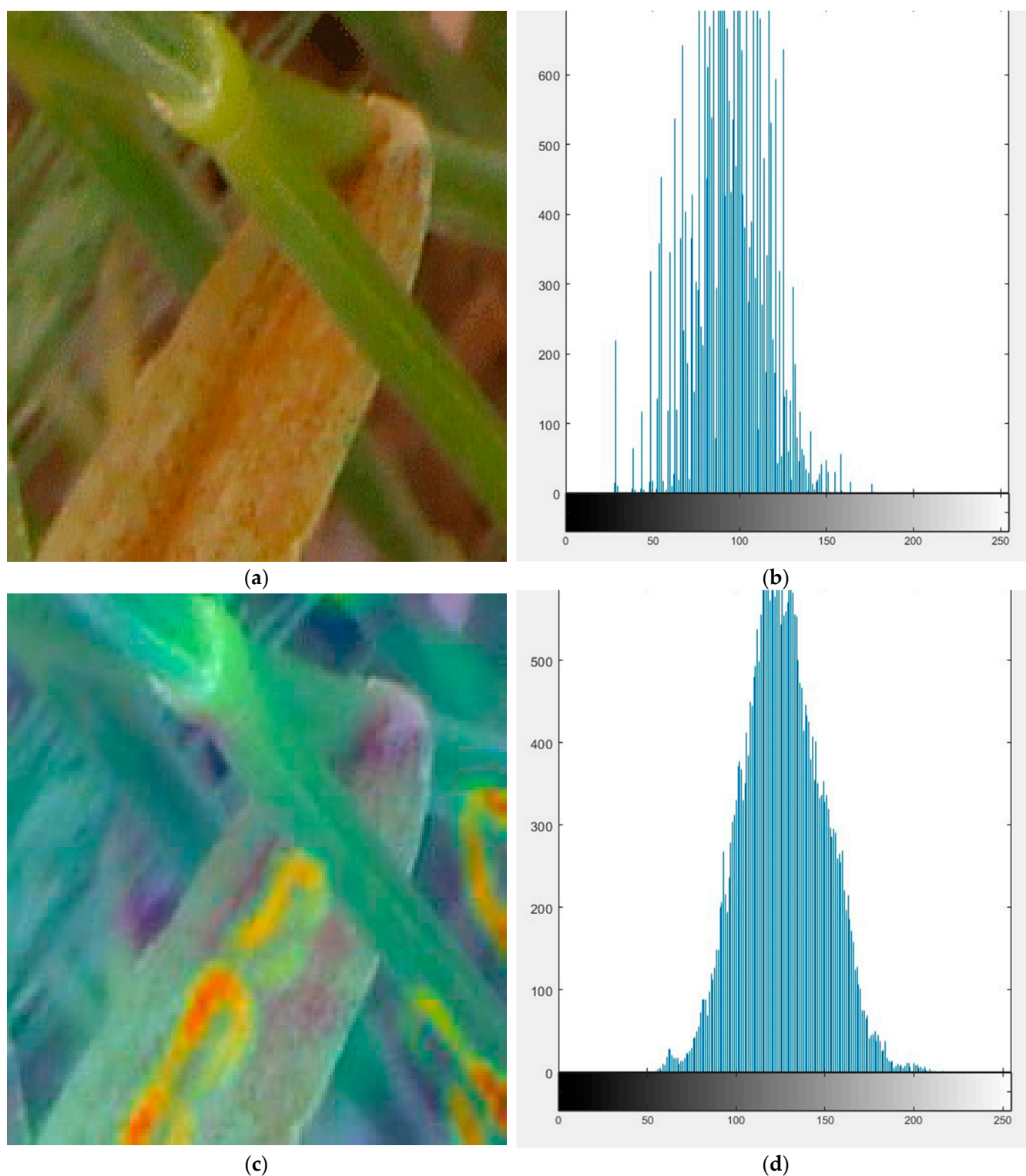


Figure 12. (a) Diseased wheat image, (b) its corresponding histogram, (c) GRADCAM image of that diseased wheat image, and (d) corresponding histogram of that GRADCAM image.

4. Discussion

The diagnosis of crop diseases at an early stage is crucial for effectively controlling and reducing crop losses. Globally, crop diseases result in significant yield reductions. For example, wheat rust disease alone leads to annual losses of over 15 million tons, valued at approximately USD 2.9 billion [29,30]. This not only fosters agricultural sustainability but also restricts the transmission of diseases. Utilizing deep learning in the classification

and detection of plant diseases provides a more effective, precise, and scalable method for managing crop health. By employing sophisticated neural network architectures, such as CNNs, deep learning systems examine images of plant leaves or crops to identify and categorize different plant diseases, including those of potato, tomato, corn, apple, and wheat [13,17]. While deep learning models (i.e., CNNs) have proven effective in analyzing plant images to detect diseases, the need for extensive data to train these models can lead to privacy issues. Due to deep learning's privacy and security flaws, it is now possible to interpret sensitive training data and steal or reverse-engineer a model. Furthermore, recent research has shown that the deep learning model is susceptible to adversarial examples that are disturbed by barely perceptible noise, which can cause the model to make highly confidently incorrect predictions [31]. In scenarios where images and data are collected directly from farms, there is a risk that confidential information about the farm's location, crop types, and farming practices might inadvertently be included or inferred from the dataset. If these data are stored or processed in external servers or cloud systems, it can increase the risk of unauthorized access or data breaches. Additionally, sharing this information with third parties, such as technology developers or researchers, without proper anonymization and consent protocols, can further compromise farm confidentiality. This research introduces a novel approach to disease classification that integrates differential privacy (DP) into deep learning models to detect fungal diseases in three crop species without compromising farm confidentiality.

The integration of DP with deep learning models has gained traction in recent years, primarily due to its ability to protect sensitive data while enabling the creation of accurate and efficient models. This approach has been successfully applied in sectors such as healthcare, banking, and social media, where it protects sensitive information from unauthorized access. In healthcare, for example, DP techniques have been used to secure patient data in electronic health records (EHRs), allowing for secure data analysis and pattern recognition without compromising individual privacy [32]. Similarly, in the financial sector, DP has facilitated the analysis of transaction data for fraud detection and risk assessment, contributing to the development of new financial products and services [33]. Social media platforms have also adopted DP to safeguard user privacy while allowing for data analysis to enhance user experience and targeted advertising [34,35].

While deep learning models are often considered "black boxes", interpretable models allow researchers to understand the factors influencing model outcomes. Although GradCAM has proven significant in various fields such as medicine [36,37], finance [38], and social science, its application in crop disease identification still remains relatively young. Covering different studies of plant and crop disease detection reveals that incorporation of deep learning models with GradCAM visualization can bridge the critical gap in our understanding of crop disease diagnostics with its unique ability to be highly class-discriminative [39]. In addition, identifying areas where the model focus on and what it overlooks can address the further training and refinement of the corresponding model with the help of heatmaps generated by GradCAM [40]. Following that, we employed GradCAM in our study to elucidate the regions of interest on leaf scans that indicate disease presence, providing valuable insights for real-time farming operations.

Our study was significantly motivated by the diverse applications of DP in these sectors, recognizing a gap in its application within agriculture. To the best of our knowledge, this is the first study to combine CNN architecture with differential privacy to eliminate privacy concerns in the agriculture sector. CropsDisNet, our proposed model, is a pioneering effort in this direction and is more competent than any other popular existing methodology. It combines deep learning with DP, addressing key concerns in agriculture such as the confidentiality of personal privacy, farming practices, soil quality, and other proprietary

information. The importance of this integration cannot be overstated, as large-scale data collection in agriculture can expose farmers and agricultural technology providers (ATPs) to numerous privacy risks, including identity theft, reputational damage, and unauthorized data access. Breaches in data privacy can have serious implications, such as the loss of intellectual property and financial growth, and may deter farmers from sharing crucial data, thereby limiting innovation and technological advancements in agriculture.

5. Conclusions

Crop diseases have been one of the main issues affecting the agricultural industry and contributing to food shortages. Making an early diagnosis is essential, as human capacity is limited. Therefore, creating an automated deep learning-based disease diagnosis model using leaf scans could aid with possible detection for early recognition and prevention. Since deep learning models are considered “black boxes”, researchers tend to maximize interpretability to better understand them. This paper proposes CropsDisNet, a novel deep learning-based model that can preserve differential privacy and also minimize privacy loss. Our knowledge of CropsDisNet’s differentiation between the healthy and unhealthy classes of three different crops (potato, corn, and wheat) was strengthened by GradCAM’s interpretability, which has so far been validated. Future studies on this research will require the involvement of larger datasets of several diseases based on leaf scans, where a differential private gradient descent method will definitely enable us to develop a viable diagnostic tool for identifying crop diseases. Future work aims to address these challenges by incorporating more extensive datasets covering a broader spectrum of diseased crops, such as rice and tomatoes. We also plan to extend our model’s applicability to other imaging domains, to develop a universal privacy-preserving deep learning model as an efficacious classification tool with robust interpretability for agricultural deployment. In addition, a comparative analysis on the performance of GradCAM with other interpretable methods such as LIME and SHAP will be considered in the future to explore the advantages and shortcomings of Grad-CAM in revealing the decision basis of the model, as well as how to combine multiple interpretable methods for a more comprehensive understanding of the model’s decision-making process.

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